

Group 2: Multivariate statistical analysis of hydrochemical and radiological data of Serbian spa waters

STAT394

This group will work with the data analysed by Tanasković et al. (2012).

Your report must contain, at least, the following elements with proper analyses and justification:

- A high-level description of the study and its motivation.
- A quantitative descriptive analysis of the data.
- A graphical/qualitative analysis of the data.
- Can you assume that the data follow a multivariate normal distribution?
- Are there “surprising” observations?

You must try to answer the questions:

- Can you improve the paper's results using Factor Analysis in terms of model quality with parsimony?
- Are the groups identified by the dendrogram consistent with changes in the linkage and distance?

References

Tanasković, I., Golobocanin, D. & Miljević, N. (2012), 'Multivariate statistical analysis of hydrochemical and radiological data of Serbian spa waters', *Journal of Geochemical Exploration* **112**, 226–234.